



BSCS 3213 / BSCS 3213L (Software Engineering)

Orientation



LET'S BEGIN IN...

05:00



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Orientation



House Rules

Attendance, Seating Arrangement, NO PHONES during class



Course Description

- Software Engineering is an introductory course that explores the fundamental principles and best practices of software development. Students will learn about the software development life cycle, including requirements elicitation and analysis, design, implementation, testing, and deployment. The course emphasizes practical application of software engineering concepts through team-based projects, where students will work collaboratively to design, build, and test software systems.

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Academic Requirements

Quizzes, Assignments, Graded Recitation, Group Project



Course Topics

Week #	Topic	Activity	Requirement
1	Orientation		
2	Introduction to Project Management		
3	Requirements Gathering and Analysis		
4	Use Case Development and Diagraming	Data Gathering	
5	Software Design and Architectural Design (UML Diagrams)	Business Case/ Research Development	Business Case / Research Proposal
6	PRELIMINARY EXAMINATION		
7	Coding Standards and Code Reviews	System Design	Logical Diagrams
8	Unit Testing and Test-Driven Development	System Design	
9	Integration Testing and Continuous Integration	System Development	Progress Reports
10	User Interface (UI) Design and Implementation	System Development	Progress Reports
11	System Testing and User Acceptance Testing (UAT)	System Development	Progress Reports
12	MIDTERM EXAMINATION		

Course Topics

Week #	Topic	Activity	Requirement
13	Project Documentation and Reporting	System Development	Progress Reports
14	System Checking	System Development	Progress Reports
15	System Checking	System Development	Progress Reports
16	System Checking	System Development	Progress Reports
17	System Checking	System Development	Progress Reports
18	FINAL EXAMINATION / PROJECT PRESENTATION		

Group Project

- Minimum of three (3) and maximum of four (4) members.
- Application and Evaluation of AI and/or data analytics in an information system.
- Documentation Requirements:
 - Approved Business Case / Research Proposal
 - Research Documentation
 - Publishable Research Format
- Final Presentation: Research Defense





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MIT 004 (Technology and Management)

Introduction to Project Management



INTENDED LEARNING OUTCOMES

- Describe what is a project and project management.
- Examine how the project management phases are conducted.
- Create a business case for a research project.

What is a Project?



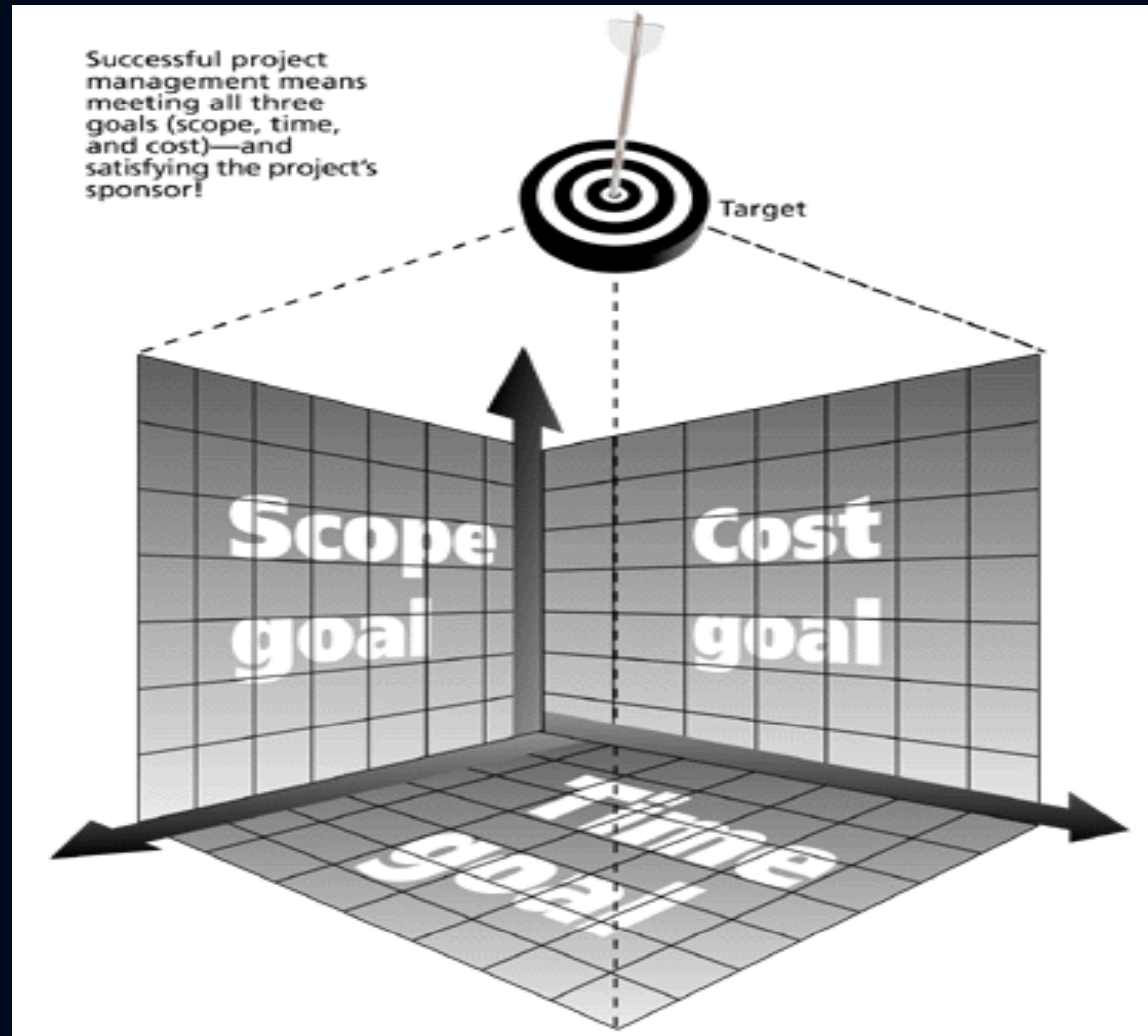
WHAT IS A PROJECT?

- A **project** is “a temporary endeavor undertaken to accomplish a unique product or service” *(PMBOK® Guide 2000, p. 4)*
- Attributes of projects
 - unique purpose
 - temporary
 - require resources, often from various areas
 - should have a primary sponsor and/or customer
 - involve uncertainty

THE TRIPLE CONSTRAINT

- Every project is constrained in different ways by its:
 - Scope goals: What is the project trying to accomplish?
 - Time goals: How long should it take to complete?
 - Cost goals: What should it cost?
- It is the project manager's duty to balance these three often competing goals

THE TRIPLE CONSTRAINT OF PROJECT MANAGEMENT



What is a Project management?



WHAT IS PROJECT MANAGEMENT?

Project management is “the application of knowledge, skills, tools, and techniques to project activities in order to meet project requirements.”

PROJECT STAKEHOLDERS

- Stakeholders are the people involved in or affected by project activities
- Stakeholders include
 - the project sponsor and project team
 - support staff
 - customers
 - users
 - suppliers
 - opponents to the project

ADVANTAGES OF USING FORMAL PROJECT MANAGEMENT

- Better control of the general financial, physical, and human resources
- Improved customer relations
- Shorter development times
- Lower costs
- Higher quality and increased reliability
- Higher profit margins
- Improved productivity
- Better internal coordination
- Higher worker morale

9 PROJECT MANAGEMENT KNOWLEDGE AREAS

- Knowledge areas describe the key competencies that project managers must develop the following:
 - 4 core knowledge areas lead to specific project objectives (scope, time, cost, and quality)
 - 4 facilitating knowledge areas are the means through which the project objectives are achieved (human resources, communication, risk, and procurement management)
 - 1 knowledge area (project integration management) affects and is affected by all of the other knowledge areas

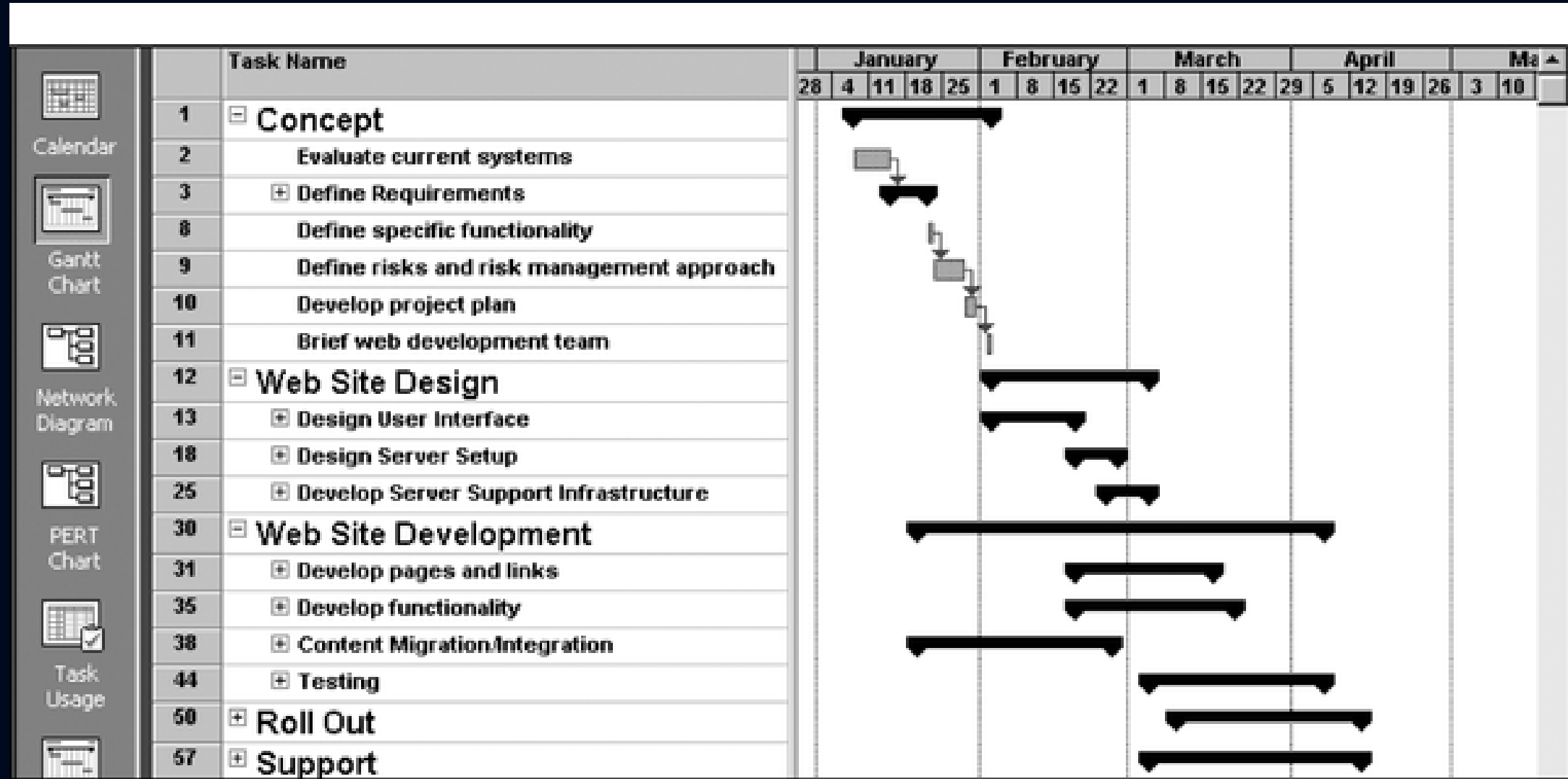
PROJECT MANAGEMENT TOOLS AND TECHNIQUES

- Project management tools and techniques assist project managers and their teams in various aspects of project management
- Some specific ones include
 - Project Charter, scope statement, and WBS (work breakdown structure) (scope)
 - Gantt charts, network diagrams, critical path analysis, critical chain scheduling (time)
 - Cost estimates and earned value management (cost)

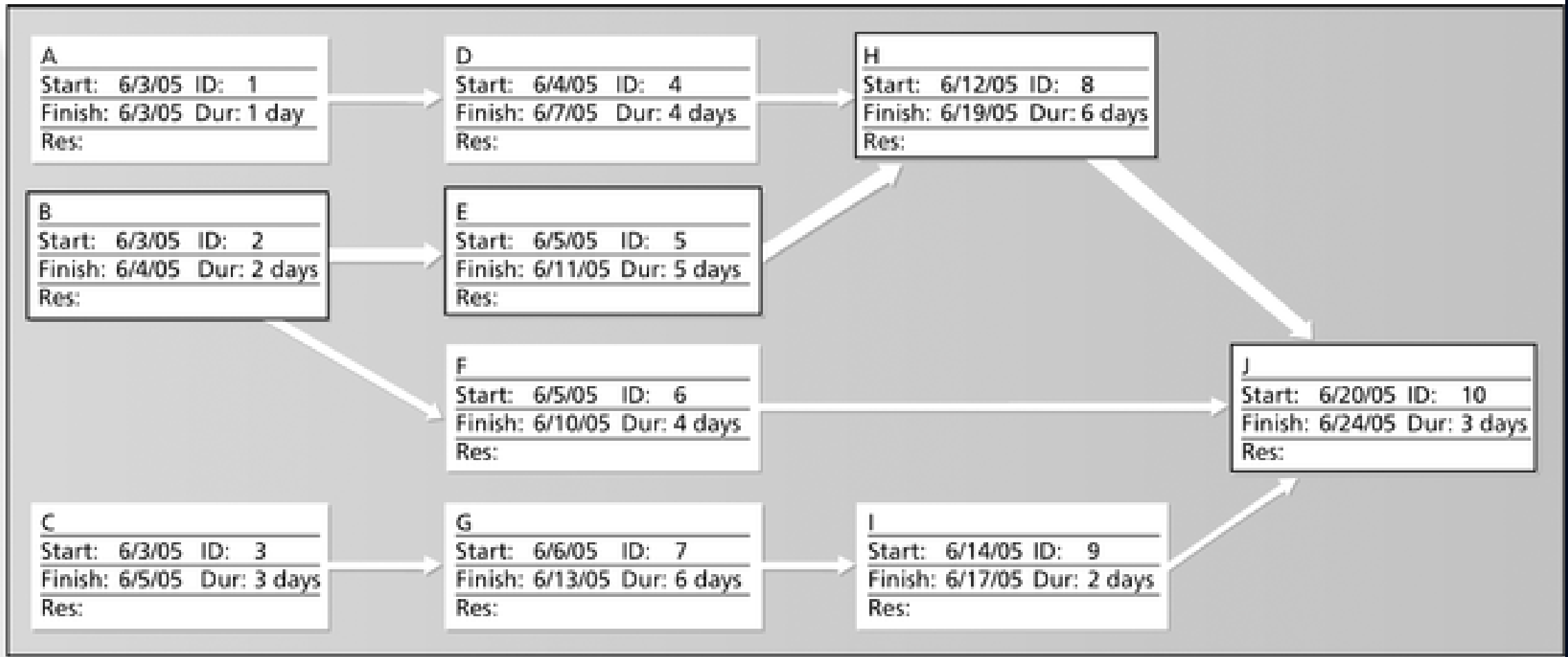
HOW PROJECT MANAGEMENT RELATES TO OTHER DISCIPLINES

- Much of the knowledge needed to manage projects is unique to the discipline of project management
- Project managers must also have knowledge and experience in
 - general management
 - the application area of the project

SAMPLE GANTT CHART



SAMPLE NETWORK DIAGRAM



What are Project Management Phases?



DEVELOPING AN IT PROJECT MANAGEMENT METHODOLOGY

- Just as projects are unique, so are approaches to project management
- Many organizations develop their own project management methodologies, especially for IT projects
- A **methodology** describes *how* things should be done; a **standard** describes *what* should be done

PROJECT PRE-INITIATION

- It is good practice to lay the groundwork for a project before it officially starts
- Senior managers often perform several pre-initiation tasks, including the following:
 - Determine the scope, time, and cost constraints for the project
 - Identify the project sponsor
 - Select the project manager
 - Meet with the project manager to review the process and expectations for managing the project
 - Determine if the project should be divided into two or more smaller projects

PROJECT INITIATION

- Initiating a project includes recognizing and starting a new project or project phase
- The main goal is to formally select and start off projects.

STAKEHOLDER REGISTER

Name	Position	Internal/ External	Project Role	Contact Information
Joe Fleming	CEO	Internal	Sponsor	joe_fleming@jwdconsulting.com
Erica Bell	PMO Director	Internal	Project manager	erica_bell@jwdconsulting.com
Michael Chen	Team member	Internal	Team member	michael_chen@jwdconsulting.com
Kim Phuong	Business analyst	External	Advisor	kim_phuong@client1.com
Louise Mills	PR Director	Internal	Advisor	louise_mills@jwdconsulting.com

PROJECT CHARTERS, BUSINESS CASES

- Charters are normally short and include key project information and stakeholder signatures
- Business Cases are the overview of what the project is.

PROJECT PLANNING

- The main purpose of project planning is to *guide execution*
- Key outputs included include:
 - A team contract
 - A project scope statement
 - A work breakdown structure (WBS)
 - A project schedule, in the form of a Gantt chart with all dependencies and resources entered
 - A list of prioritized risks (part of a risk register)

PROJECT EXECUTING

- Usually takes the most time and resources to perform project execution
- Project managers must use their leadership skills to handle the many challenges that occur during project execution

PROJECT MONITORING AND CONTROLLING

- Involves measuring progress toward project objectives, monitoring deviation from the plan, and taking correction actions
- Affects all other process groups and occurs during all phases of the project life cycle
- Outputs include performance reports, change requests, and updates to various plans

PROJECT CLOSING

- Involves gaining stakeholder and customer acceptance of the final products and services
- Even if projects are not completed, they should be closed out to learn from the past
- Outputs include project files and lessons-learned reports, part of organizational process assets
- Most projects also include a final report and presentation to the sponsor/senior management

PROJECT MONITORING TOOLS

- Asana
- Trello
- Monday.com
- ClickUp
- Wrike
- Jira

Questions?





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